ModelIt!

Teacher's Guide

# How Do Rocks Tell Stories?

Reading Earth's History from Rocks and Layers

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 2 |
| **NGSS Standard** | 2-ESS1-1 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: 2-ESS1-1**

Use information from several sources to provide evidence that Earth events can occur quickly or slowly.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Obtaining, Evaluating, and Communicating Information** — Students use information from rocks and layers to provide evidence about Earth events and whether they occurred quickly or slowly. |
| **Disciplinary Core Idea** | **ESS1.C The History of Planet Earth** — Some events happen very quickly; others occur very slowly, over a time period much longer than one can observe. Rock layers and fossils provide evidence about Earth's history. |
| **Crosscutting Concept** | **Stability and Change** — Students explore how some Earth processes are slow and steady (erosion) while others are sudden and dramatic (volcanic eruptions), and how both leave evidence in rocks. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A scientist finds smooth, round rocks at the bottom of a valley. What Earth event most likely shaped these rocks?

**CONSTRUCTED RESPONSE:**

Explain how rock layers help scientists figure out what happened on Earth long ago. How can you tell which layer is the oldest?

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Rocks are the oldest storytellers on Earth. Long before humans existed, rocks were being formed, broken down, and reformed — creating a record of Earth's history that scientists can read like a book.

## Rocks Tell Stories

Every rock has a story about how it formed. Igneous rocks formed from hot melted rock (lava or magma) that cooled and hardened — often from volcanic eruptions. Sedimentary rocks formed when tiny pieces of rock, sand, and dirt (called sediment) settled in layers and were pressed together over millions of years. Metamorphic rocks formed when existing rocks were changed by heat and pressure deep underground. Each type gives us clues about what happened in the past.

## Reading Rock Layers

When sediment settles, it forms layers — like pancakes stacking up over time. The bottom layer is the oldest because it was deposited first. Each new layer on top is younger. Scientists called geologists read these layers like chapters in a history book. Different colored or textured layers show different time periods and different events. A dark layer might be from a volcanic eruption. A sandy layer might be from an ancient beach.

## Fast and Slow Earth Events

Some Earth events happen in seconds or minutes — earthquakes shake the ground, volcanoes erupt with explosive force, and landslides send rock tumbling down mountains. Other events take thousands or millions of years — rivers slowly carve canyons, wind gradually sculpts rock formations, and sediment builds up one grain at a time. Both fast and slow events leave evidence in rocks that scientists can study today.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify Earth event as an external component and rock type and rock layer pattern as internal components that form differently depending on what event occurred.

**E** **Establish Relationships**
Students determine that different Earth events produce different rock types — slow events create smooth layered rocks, while fast events create rough textured rocks.

**V** **Visualize & Model**
Students build a model showing how different Earth events (river, volcano, wind) create different rock types and layer patterns.

**E** **Evaluate Outcomes**
Students run slow river, fast volcano, and slow wind scenarios to compare the rock types and layers each event produces.

**R** **Revise & Extend**
Students add fossils in layers or weathering speed to explore more of what rock layers can tell us about the past.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

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|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"What if I told you that rocks can tell STORIES? Not with words — but with clues hidden inside them!"*

**Watch For**

Pass around 2-3 different rocks (smooth river rock, rough volcanic rock, layered sedimentary rock). Ask: How are these different?

## Slide 2: What Will We Learn Today?

3 min

**Say:** *"Today we are going to learn to read rocks like books — each one tells a story about Earth's past!"*

**Watch For**

Read objectives together. Introduce rock layer, erosion, and sediment with real examples.

## Slide 3: The Big Question

3 min

**Say:** *"These rocks are millions of years old. They were here before people, before cities, even before most animals! What stories are they hiding?"*

**Watch For**

Show a photo of the Grand Canyon. Ask: Do you see the layers? Each layer is a different chapter in Earth's story.

## Slide 4: Let's Build a Model!

3 min

**Say:** *"We are going to build a model to understand how different events on Earth create different rocks."*

**Watch For**

Preview LEVER steps. Demo: shake a jar of water with sand and gravel — watch the layers settle. That is how sediment works!

## Slide 5: Activity 1: Sort the Parts

7 min

**Say:** *"What are the parts of our rock story system? An Earth event happens. Then a certain type of rock forms. Then we see it in the layers."*

**Watch For**

Guide sorting: Earth Event is external (nature does it). Rock Type and Rock Layer Pattern are internal (they result from the event).

## Slide 6: Activity 2: Connect with Arrows

7 min

**Say:** *"When a volcano erupts, does it make the same kind of rock as a gentle river? Let us connect our parts!"*

**Watch For**

Draw arrows from Earth Event to Rock Type and Rock Layer Pattern. Discuss how volcanoes and rivers make DIFFERENT rocks.

## Slide 7: Activity 3: Run the Simulation!

8 min

**Say:** *"Let us test a slow river and a fast volcano! What kind of rock does each one make?"*

**Watch For**

Students predict, then run simulations. Compare smooth river rocks to rough volcanic rocks. Discuss fast vs. slow events.

## Slide 8: What Did We Discover?

7 min

**Say:** *"Now you can read rocks! Smooth rocks tell slow stories. Rough rocks tell fast, exciting stories. And rock layers are like chapters in Earth's history book!"*

**Watch For**

Each student gets a rock sample and writes one sentence about what story it might tell. Share with a partner.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Earth Event
- **INTERNAL:** Rock Type, Rock Layer Pattern

## Relationships

- **Earth Event (slow) to Rock Type:** POSITIVE (+) — Slow, gentle Earth events like rivers flowing for millions of years create smooth, layered sedimentary rocks. The longer and gentler the event, the smoother and more organized the rock layers become.
- **Earth Event (fast) to Rock Type:** POSITIVE (+) — Fast, powerful Earth events like volcanic eruptions create rough, bubbly igneous rocks. The hotter and faster the event, the more dramatic and textured the resulting rock.

## Key Discoveries

- Different Earth events create different types of rocks — volcanoes make rough, bubbly rock, rivers make smooth, rounded rock
- Rock layers are like pages in a history book — the bottom layers are oldest and the top layers are newest
- Some Earth events happen quickly (earthquakes, volcanoes) and some happen slowly (erosion, sediment building up)
- Scientists read rock layers to figure out what happened on Earth millions of years ago

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "All rocks are the same" | Rocks are incredibly different from each other! Smooth river rocks, rough volcanic rocks, sparkly mineral rocks, and layered sedimentary rocks all formed in completely different ways. The type of rock tells us what Earth event created it — like a clue about the past. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Rocks have always been there and never change" | Rocks are constantly being created and destroyed! Volcanoes make new rock from lava. Rivers slowly wear rock down into sand. Sand gets pressed into new rock over millions of years. Rocks are always changing — it just happens so slowly we usually cannot see it. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "The top layer of rock is the oldest" | Actually, the BOTTOM layer is the oldest! Layers build up from the bottom, one on top of another. The bottom layer was there first. Each new layer settles on top. So if you dig down into the Earth, you are going BACK in time. The deepest rocks tell the oldest stories. |